Function-al vectors

Some fourier analysis and linear algebra

Vivek Kalita

Department of Physics, IISER Mohali

17 October 2026

Contents

Some basic Linear Algebra 1

Exploring functions as bases 2

2.1 The trignometric basis 3

2.2 When can you fourier-expand? 3

2.3 The complex exponential as the basis 6

1 Some basic Linear Algebra

Before going into the main topic, lets first start with a basic review.

Vector spaces (or more expressively linear spaces) are defined by two operations: addition and scalar multiplication.

Definition 1.1 (Vector Space).

Let $\mathbb{F}$ be a field. A **vector space over** $\mathbb{F}$ is an abelian group $(V, +)$ equipped with a scalar multiplication

$$\mathbb{F} \times V \longrightarrow V, \qquad (a, v) \longmapsto av,$$

such that, for all $a, b \in \mathbb{F}$ and $u, v \in V$,

1. $a(u + v) = au + av$,
2. $(a + b)v = av + bv$,
3. $(ab)v = a(bv)$,
4. $1_{\mathbb{F}} v = v$.

Now consider two vectors $\mathbf{u}$ and $\mathbf{v}$, it is easy to see that the space given by:

$$V = \{a\mathbf{u} + b\mathbf{v} \,|\, a, b \in \mathbb{F}\}$$

could generate a subspace of the vector space, such that it is a vector space in itself.

In general, for any vector space there exist a set of vectors (the basis) whose linear combinations would produce the entire vector space.

As for example,

1. In $\mathbb{R}^3$, you can produce all the vectors with the simple basis $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$.

Definition 1.2 (Basis).

Let V be a vector space over a field $\mathbb{F}$. A subset $B = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \subseteq V$ is called a *basis* of V if:

1. B spans V, i.e., every $\mathbf{v} \in V$ can be expressed as
$$\mathbf{v} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n, \qquad a_i \in \mathbb{F}.$$
2. B is linearly independent, i.e.,
$$a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n = \mathbf{0}$$
implies
$$a_1 = a_2 = \cdots = a_n = 0.$$

Now till now what we've done is algebra. To induce geometry into the theory of linear spaces, we need to induce a metric.

We do this by introducing the inner product, which consequenly allows us to define the metric and also the angular measure between two vectors.

Definition 1.3 (Inner Product).

Let V be a vector space over $\mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. An inner product on V is a map
$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{F}$$
satisfying the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{F}$:

1. Linearity
$$\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle.$$
2. Conjugate symmetry
$$\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}.$$
3. Positive definiteness
$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0,$$
with equality if and only if $\mathbf{v} = \mathbf{0}$.

The **norm** of a vector $\mathbf{v}$ consequently is given as:
$$\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle.$$

Consequently, we can also define the angle between two non-zero vectors by
$$\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

We could thus define a notion of a orthogonal basis.

Definition 1.4 (Orthogonal Basis).

Let V be an inner product space over $\mathbb{F}$. A basis $B = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ of V is called an *orthogonal basis* if its distinct elements are mutually orthogonal, i.e.,
$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = 0, \qquad i \neq j.$$
If, in addition, every basis vector has unit norm,
$$\|\mathbf{e}_i\| = 1, \qquad i = 1, 2, \ldots, n,$$
the basis is called an *orthonormal basis*.

2 Exploring functions as bases

Consider the follwing integrals:
$$\int_{-\pi}^{\pi} \cos(nx)\cos(mx)\,dx = \pi\delta_{nm}, \qquad \int_{-\pi}^{\pi} \sin(nx)\sin(mx)\,dx = \pi\delta_{nm}, \qquad \int_{-\pi}^{\pi} \sin(nx)\cos(mx)\,dx = 0.$$

Quite easily you can see where I am trying to get at. The functions

$$\mathcal{B} = \{1, \frac{1}{\pi}\cos(nx), \frac{1}{\pi}\sin(nx)\}$$

could potentially be considered as the basis of a a vector space.

2.1 The trignometric basis

With the basis laid over, we can define a vector space with $\mathcal{B}$ as the basis:

Definition 2.1 (Space of Trigonometric Polynomials).

Let $\mathcal{T}$ be a vector space over $\mathbb{R}$, spanned by the set of functions

$$\mathcal{B} = \left\{1, \frac{1}{\pi}\cos(nx), \frac{1}{\pi}\sin(nx) \;\middle|\; n \in \mathbb{N}\right\}.$$

Thus, every $f \in \mathcal{T}$ can be expressed as

$$f(x) = a_0 + \sum_{n=1}^{N}\left[\frac{a_n}{\pi}\cos(nx) + \frac{b_n}{\pi}\sin(nx)\right],$$

where $a_n, b_n \in \mathbb{R}$ and N is finite. We equip $\mathcal{T}$ with the inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,dx, \qquad f, g \in \mathcal{T}.$$

In general, given the function $f(x)$, we can also extract any of the coefficients. And also we have some very basic results:

Theorem 2.2 (Basic Results).

Let $f \in \mathcal{T}$ have the representation

$$f(x) = a_0 + \sum_{n=1}^{N}\left[\frac{a_n}{\pi}\cos(nx) + \frac{b_n}{\pi}\sin(nx)\right].$$

Then the following results hold:

1. Fourier Coefficients: *The coefficients are given by*

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \quad b_n = \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx.$$

2. Parseval's Identity: *The squared norm of f is given by*

$$\|f\|^2 = \langle f, f\rangle = \int_{-\pi}^{\pi} f^2(x)\,dx = 2\pi a_0^2 + \frac{1}{\pi}\sum_{n=1}^{N}(a_n^2 + b_n^2).$$

2.2 When can you fourier-expand?

With our choice of basis $\mathcal{B}$, the Fourier coefficients are given by

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx, \qquad a_n = \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad b_n = \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx.$$

For all these integrals to exist, it is sufficient that $f \in L^1([-\pi, \pi])$, which means that $f(x)$ is absolutely integrable over the interval, i.e.,

$$\int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

To see why this condition is sufficient, observe that

$$|\cos(nx)| \leq 1, \qquad |\sin(nx)| \leq 1.$$

Consequently,

$$\left|\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx\right| \le \int_{-\pi}^{\pi} |f(x)\cos(nx)|\,dx$$
$$\le \int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

The same argument applies to the sine coefficients. Thus, all the Fourier coefficients are well-defined.

Theorem 2.3 (Existence of Fourier Coefficients).

The Fourier coefficients of a function f are given by

$$A_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx, \qquad A_n = \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad B_n = \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx,$$

provided that f is absolutely integrable over the interval $[-\pi,\pi]$. That is:

$$\int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

Proof.

1. ($\Rightarrow$) Suppose all the Fourier coefficients exist as finite integrals. In particular, A_0 exists, so

   $$\int_{-\pi}^{\pi} f(x)\,dx$$

   is finite. Hence f is absolutely integrable, i.e.,

   $$\int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

2. ($\Leftarrow$) Conversely, suppose that

   $$\int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

   Since

   $$|\cos(nx)| \le 1, \qquad |\sin(nx)| \le 1,$$

   we have

   $$\int_{-\pi}^{\pi} |f(x)\cos(nx)|\,dx \le \int_{-\pi}^{\pi} |f(x)|\,dx < \infty,$$
   $$\int_{-\pi}^{\pi} |f(x)\sin(nx)|\,dx \le \int_{-\pi}^{\pi} |f(x)|\,dx < \infty.$$

   Thus, all the Fourier coefficients exist as finite integrals.

□

Example 2.4

But now, consider the function

$$f(x) = \frac{1}{\sqrt{|x|}}, \qquad x \in [-\pi, \pi], \quad x \neq 0.$$

1. If we define $f(0) = 0$, then we can easily verify that

$$\int_{-\pi}^{\pi} |f(x)|\, dx = 4\sqrt{\pi} < \infty.$$

   Thus, $f \in L^1([-\pi, \pi])$.

2. However, it is easily seen that

$$|f(x)|^2 = \frac{1}{|x|},$$

   and hence

$$\int_{-\pi}^{\pi} |f(x)|^2\, dx = 2\int_{0}^{\pi} \frac{1}{x}\, dx = \infty.$$

   Therefore, $f \notin L^2([-\pi, \pi])$.

Essentially, this demonstrates that absolute integrability alone is not sufficient to guarantee a finite norm. Consequently, if we wish to equip our function space with the inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x)\, dx,$$

we must restrict ourselves to square-integrable functions. Thus, our function space should be $L^2([-\pi, \pi])$, which ensures that every function has a finite norm and that its Fourier coefficients are well-defined.

With this final restriction, we now have a complete description of the trignometric space.

Theorem 2.5 (Fourier Expansion in L^2).

The trigonometric system

$$\mathcal{B} = \left\{ 1, \frac{1}{\pi}\cos(nx), \frac{1}{\pi}\sin(nx) \;\middle|\; n \in \mathbb{N} \right\}$$

forms a complete orthogonal basis of $L^2([-\pi, \pi])$. Consequently, every $f \in L^2([-\pi, \pi])$ admits a Fourier expansion

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[\frac{a_n}{\pi}\cos(nx) + \frac{b_n}{\pi}\sin(nx) \right],$$

where the coefficients are given by the corresponding Fourier integrals, and the partial sums converge to f in the L^2 norm.

The formal proof of this would require piecing a few more parts. And hence would be discussed later.

There is yet another very interesting thing to observe about the fourier coefficients. The zeros of $\cos(nx)$ are located at

$$x_k = \frac{(2k+1)\pi}{2n}, \qquad k \in \mathbb{Z}.$$

The separation between two consecutive zeros is

$$\Delta x = \frac{\pi}{n} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Approximate $f(x)$ by a step function, taking a constant value c_j over each interval $[p_j, q_j]$. Then

$$b_n = \int_{-\pi}^{\pi} f(x)\cos(nx)\, dx \approx \sum_{j=1}^{m} c_j \int_{p_j}^{q_j} \cos(nx)\, dx.$$

Since

$$\left| \int_{p_j}^{q_j} \cos(nx)\, dx \right| \leq \frac{2}{n},$$

we obtain

$$|b_n| \lesssim \frac{2}{n} \sum_{j=1}^{m} |c_j| \longrightarrow 0 \qquad \text{as } n \to \infty.$$

A similar argument holds for the other Fourier coefficient.

Theorem 2.6 (Vanishing of Fourier Coefficients).
Let $f \in L^1([-\pi, \pi])$ have the Fourier expansion

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[\frac{a_n}{\pi} \cos(nx) + \frac{b_n}{\pi} \sin(nx) \right].$$

Then the Fourier coefficients satisfy

$$\lim_{n\to\infty} a_n = 0, \qquad \lim_{n\to\infty} b_n = 0.$$

2.3 The complex exponential as the basis

Consider the following integral:

$$\int_{-\pi}^{\pi} e^{imx} e^{-inx}\, dx = 2\pi\delta_{mn}.$$

This isn't suprising, as the complex exponential inherently contains the trignometric terms. In general you can define the vector space:

Definition 2.7 (Complex Fourier Series).
Let $\mathcal{H} = L^2([-\pi, \pi]; \mathbb{C})$ be the vector space of complex-valued square-integrable functions, equipped with the inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g^*(x)\, dx, \qquad f, g \in \mathcal{H}.$$

The set of complex exponential functions

$$\mathcal{B} = \left\{ e^{imx} \mid m \in \mathbb{Z} \right\}$$

forms a complete orthogonal basis of $\mathcal{H}$. Consequently, every $f \in \mathcal{H}$ admits a Fourier expansion of the form

$$f(x) = \sum_{m=-\infty}^{\infty} c_m e^{imx},$$

where the Fourier coefficients are given by

$$c_m = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-imx}\, dx.$$

We can derive some basic results from the complex representation.

1. Parseval's Identity:

$$\|f\|^2 = 2\pi \sum_{m=-\infty}^{\infty} c_m c_m^*.$$

2. Riemann-Lebesgue Lemma:
   For $f \in L^1([-\pi, \pi])$, its Fourier coefficients satisfy

$$\lim_{|m|\to\infty} c_m = 0.$$

3. Conjugate Symmetry:
   If $f(x)$ is real-valued, its Fourier coefficients satisfy

$$c_{-m} = c_m^*.$$

   and they would be related to the sine and cosine coefficient (for the real-valued fourier expansion) as:

$$a_0 = c_0, \qquad a_n = \pi(c_n + c_{-n}), \qquad b_n = i\pi(c_n - c_{-n}).$$

4. Inner Product Identity:

   For $f, g \in L^2([-\pi, \pi])$, with Fourier coefficients c_m and d_m, respectively,

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g^*(x)\, dx = 2\pi \sum_{m=-\infty}^{\infty} c_m d_m^*.$$